

Analytical Chemistry II - Named Scientists & Eponymous Methods

Collected from the uploaded PDF only. The same name is listed more than once when the file uses it in different contexts.

How to read this sheet	Name = scientist / eponymous method. What it is = the role in the file. Where mentioned = lecture/topic in the uploaded PDF. Repeated? = same name used again in another context, if applicable.
-------------------------------	---

1) Scientists / theories

Name	What it is / does	Where mentioned	Repeated? / difference
Arrhenius	Arrhenius acid-base theory: acids ionize to give H^+ and bases ionize to give OH^- . The file notes that this theory does not discuss the role of solvent.	Lecture 1 / Acid-Base Theories	Not repeated with a different meaning; only used once as a theory.
Brønsted-Lowry	Acid = proton donor; base = proton acceptor. The file also notes that the solvent can take part as acid or base, and water is amphoteric.	Lecture 1 / Acid-Base Theories	Used twice in the file: once for the theory itself and again to explain the solvent role in water.
Lewis	Acid = electron-pair acceptor; base = electron-pair donor. The file uses NH_3 and Ag^+ as examples.	Lecture 1 / Acid-Base Theories	Not repeated with a different meaning; only used once as a theory.
Ostwald	Ostwald theory of color indicators: the indicator exists as weak acid/base forms, and the undissociated form has a different color from the ionized form.	Lecture 1 / Color indicators	Not repeated with a different meaning; only used once.
Henderson (Henderson equation)	Buffer equation used for acidic and basic buffers. In the slides it appears as 'Henderson Basel Equation'. The same formula is applied to weak acid/salt and weak base/salt systems.	Lecture 2 / Buffer solutions	Repeated twice in practice: acidic buffer and basic buffer. The formula is the same, only the system changes.

2) Named titration methods

Name	What it is / does	Where mentioned	Repeated? / difference
Mohr's method	Precipitation method for halides with chromate indicator. The file says it works in neutral or slightly alkaline medium and is limited by pH and by some ions (e.g., iodide and thiocyanate give false endpoint problems).	Lecture 10 / Precipitometric titrations	Mentioned more than once: once as the method itself, and again in the comparison/limitations with Volhard and Fajans.

Name	What it is / does	Where mentioned	Repeated? / difference
Fajans' method	Adsorption-indicator method for direct estimation of halides by titration against AgNO ₃ . Uses indicators such as fluorescein, eosin, or rhodamine 6G.	Lecture 10 / Precipitometric titrations	Mentioned more than once: the method title and again in the comparison with Volhard's method.
Volhard's method	Residual (back) determination of halides with known excess AgNO ₃ in acidic medium, then back titration with NH ₄ SCN using ferric alum as indicator. The file also says it can be used for direct titration of silver salts with thiocyanate.	Lecture 10 / Precipitometric titrations	Repeated twice with a real difference: (1) back titration of halides; (2) direct titration of Ag ⁺ with thiocyanate.
Formol titration / Formol method	Indirect titration used for ammonium salts. Formaldehyde converts the ammonium salt into a product that liberates HCl, and that HCl is titrated with NaOH.	Lecture 6 / Indirect titration	Repeated once in the same section, and the file also notes the same reaction can be used for formaldehyde determination.

3) Other named method terms in the file

Name	What it is / does	Where mentioned	Repeated? / difference
EDTA	Core complexing agent in complexometric titration. The file repeatedly uses EDTA in direct, back, and indirect titration examples.	Lecture 8 / Complexometric titrations	Repeated many times because it is the main reagent, not a person-name.
Hydroxylamine hydrochloride method	Used in indirect titration for aldehydes and ketones; it liberates HCl that is then titrated with NaOH.	Lecture 6 / Indirect titration	A named reagent-method, but not an eponymous scientist name.
Biphasic titration system	Sodium salicylate is titrated with HCl in ether; salicylic acid is removed to avoid a false early endpoint.	Lecture 5 / Biphasic titration	Not a scientist name, but a named method/system in the file.

4) Same name, repeated in another context

Name	What it is / does	Where mentioned	Repeated? / difference
Brønsted-Lowry (repeated context)	The file repeats the Brønsted-Lowry idea again in the non-aqueous titration lecture to explain how solvent choice affects acid/base strength.	Lecture 6 / Non-aqueous titration	Same name, different context: aqueous theory vs solvent-dependent behavior in non-aqueous titration.
Volhard (repeated context)	In one place, Volhard is described as a back titration for halides; in another, the file says it is applied for direct titration of silver salts with thiocyanate.	Lecture 10 / Precipitometric titrations	Same name, two applications with different analyte/titrant directions.

Important file-note

The file spells one buffer equation as 'Henderson Basel Equation'. In the final sheet I kept the wording close to the file while treating it as the Henderson buffer equation used for acidic and basic buffers.